

Name: Manahil Habib

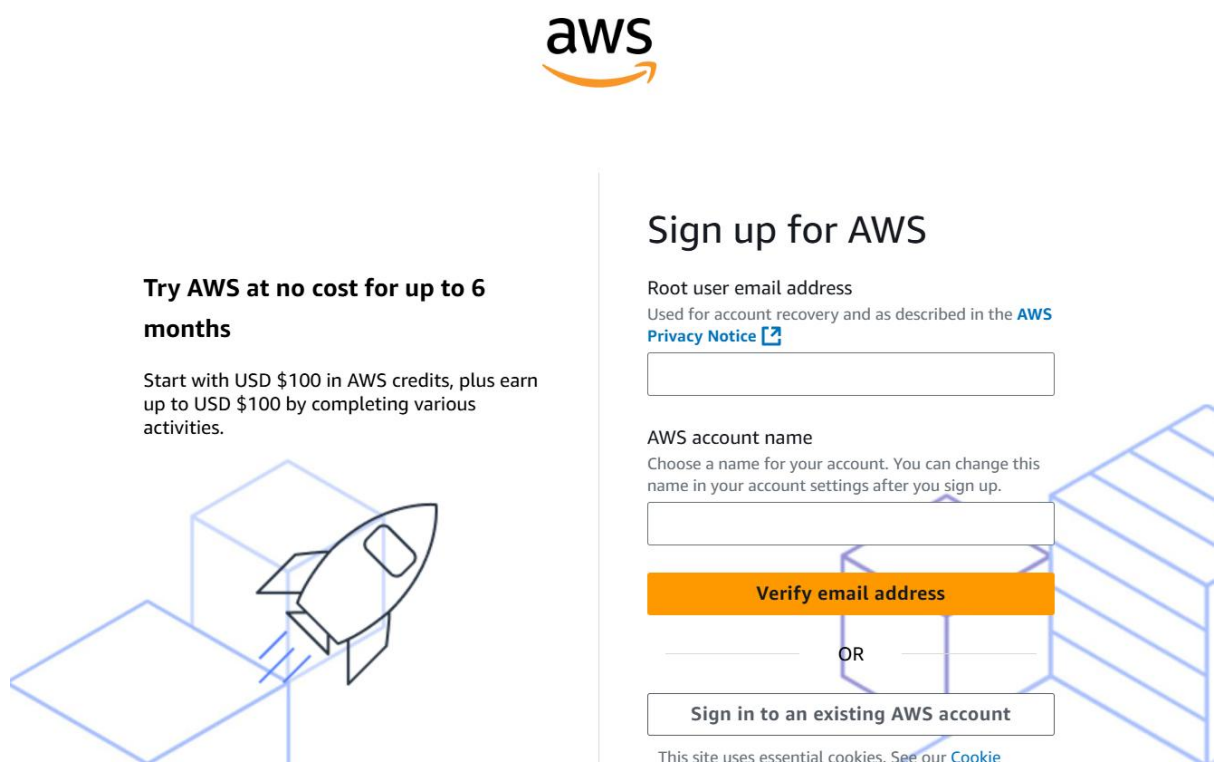
Section: 5-B

Reg no: 2023-BSE-042

Lab 8

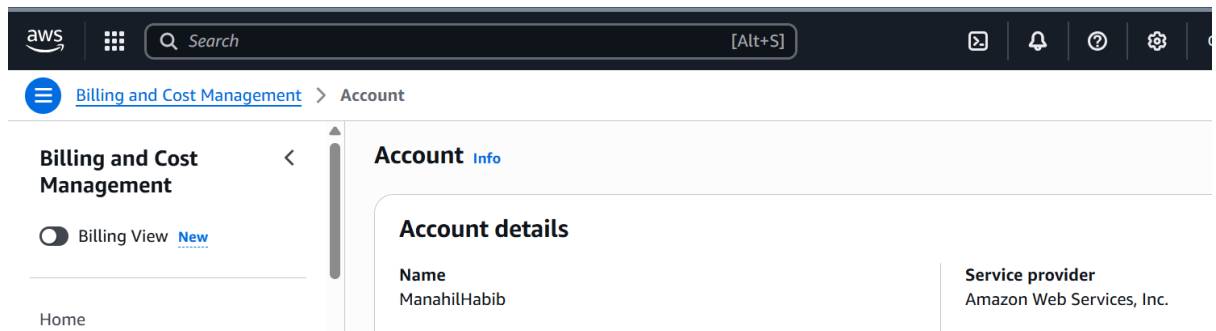
Task 1

1. Open your browser and go to: [AWS Signup](#)



The image shows the AWS Signup page. At the top is the AWS logo. Below it, on the left, is a section titled "Try AWS at no cost for up to 6 months" with a subtext: "Start with USD \$100 in AWS credits, plus earn up to USD \$100 by completing various activities." Below this text is an illustration of a rocket launching from a launch pad. On the right side of the page is the "Sign up for AWS" form. It includes a "Root user email address" field with a subtext: "Used for account recovery and as described in the [AWS Privacy Notice](#)". Below this is an "AWS account name" field with a subtext: "Choose a name for your account. You can change this name in your account settings after you sign up." Below the name field is an orange "Verify email address" button. Below the button is an "OR" separator. Below the separator is a "Sign in to an existing AWS account" button. At the bottom of the form, there is a small text: "This site uses essential cookies. See our [Cookie](#)".

2. Complete registration (Account type: Personal, Plan: AWS Paid Plan), fill contact, billing (credit card) and phone details, complete verification. After successful registration capture:



The image shows the AWS Account Dashboard. At the top is the AWS logo and a search bar. Below the logo is a navigation bar with "Billing and Cost Management" and "Account". The main content area is divided into two sections. The left section is titled "Billing and Cost Management" and has a "Billing View" button. The right section is titled "Account" and has an "Info" link. Below the "Account" link is the "Account details" section. It includes a "Name" field with the value "ManahilHabib" and a "Service provider" field with the value "Amazon Web Services, Inc.".

3. Sign in as the root user (root email). Immediately capture:

Sign In

Access your AWS account by user type.

User type [\(not sure?\)](#)

☒ **Root user**
Account owner that performs tasks requiring unrestricted access.

☐ **IAM user**
User within an account that performs daily tasks.

Email address

23-22411-042@se.fjwu.edu.pk

Next

OR

aws

Search

[Alt+S]

Europe (Stockholm)

Account ID: 0950-3239-1730

ManahilHabib

☰

Console Home [Info](#)

[Reset to default layout](#) [+ Add widgets](#)

☰ Recently visited [Info](#)

No recently visited services

Explore one of these commonly visited AWS services.

[EC2](#) [S3](#) [Aurora and RDS](#) [Lambda](#)

☰ Applications (0) [Info](#)

[Create application](#)

Region: Europe (Stockholm)

Select Region [eu-north-1 \(Current Region\)](#)

< 1 >

Name	Description	Region	Orig
No applications			
Get started by creating an application.			
Create application			

[CloudShell](#) [Feedback](#) [Console Mobile App](#)

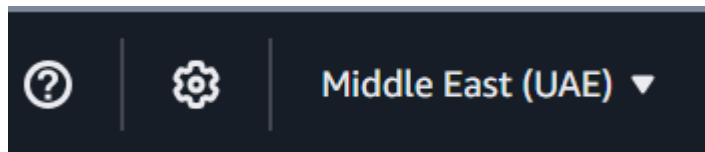
© 2025, Amazon Web Services, Inc. or its affiliates. [Privacy](#) [Terms](#) [Cookie preferences](#)

4. From the Console, open the region selector and enable UAE (me-central-1), then switch to me-central-1. Capture the change:

☐	Europe (Zurich)	✖ Disabled
☐	Europe (Milan)	✖ Disabled
☐	Europe (Spain)	✖ Disabled
☐	Israel (Tel Aviv)	✖ Disabled
☒	Middle East (UAE)	✖ Disabled
☐	Middle East (Bahrain)	✖ Disabled
☐	Mexico (Central)	✖ Disabled
☐	Asia Pacific (Tokyo)	✔ Enabled by default
☐	Asia Pacific (Seoul)	✔ Enabled by default

finishes preparing the Region, you can access it from the region selector located in the top navigation bar.

☐	Europe (Spain)	✖ Disabled
☐	Israel (Tel Aviv)	✖ Disabled
☐	Middle East (Bahrain)	✖ Disabled
☐	Mexico (Central)	✖ Disabled
☐	Middle East (UAE)	ⓘ Enabling

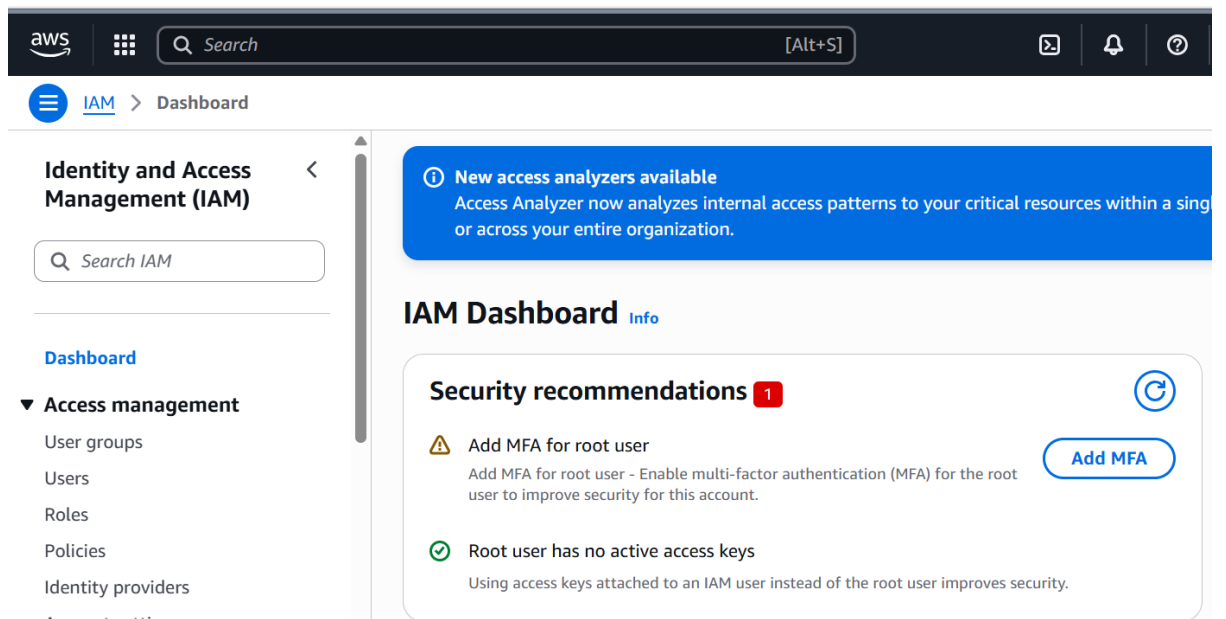


5. Task 1 summary screenshot (combine evidence):

A screenshot of the AWS Management Console home page. The top navigation bar shows the AWS logo, a search bar, and the region 'Middle East (UAE)'. The main content area has a 'Console Home' header with 'Info' and 'Reset to default layout' and 'Add widgets' buttons. Below the header, there are two main sections: 'Recently visited' and 'Applications'. The 'Recently visited' section shows 'No recently visited services' and lists 'EC2', 'S3', 'Aurora and RDS', and 'Lambda'. The 'Applications' section shows 'Region: Middle East (UAE)' and 'me-central-1 (Current Region)'. It also has a 'Find applications' search bar and a 'Create application' button.

Task 2

1. Open IAM via Console search (Alt+S → "IAM").



2. Create the Admin user: IAM → Users → Create user. Fill:

User details

User name

Admin

The user name can have up to 64 characters. Valid characters: A-Z, a-z, 0-9, and + = , . @ _ - (hyphen)

- ☒ Provide user access to the AWS Management Console - *optional*

In addition to console access, users with `SignInLocalDevelopmentAccess` permissions can use the same console credentials for programmatic access without the need for access keys.

Console password

- ☒ Autogenerated password

You can view the password after you create the user.

- ☐ Custom password

Enter a custom password for the user.

- Must be at least 8 characters long
- Must include at least three of the following mix of character types: uppercase letters (A-Z), lowercase letters (a-z), number, % ^ & * () _ + - (hyphen) = [] { } | ' "

☐ Show password

[Alt+S]

Global
ManahilHabib

Permissions options

☐ **Add user to group**
Add user to an existing group, or create a new group. We recommend using groups to manage user permissions by job function.

☐ **Copy permissions**
Copy all group memberships, attached managed policies, and inline policies from an existing user.

☒ **Attach policies directly**
Attach a managed policy directly to a user. As a best practice, we recommend attaching policies to a group instead. Then, add the user to the appropriate group.

Permissions policies (1/1434)

Choose one or more policies to attach to your new user.

Filter by Type: All types

	Policy name	Type	Attached entities
☐	AccessAnalyzerServiceRol...	AWS managed	0
☒	AdministratorAccess	AWS managed - job function	0

User created successfully

You can view and download the user's password and email instructions for signing in to the AWS Management Console.

[View user](#)

- Download the Admin .csv and show its presence on your Windows host (do not display the password text):

C11

	A
1	User name
2	Admin

- Sign out of root, then sign in using the Admin account (use the signin URL from the .csv). Capture after successful Admin login:

IAM username

Admin

Password

.....

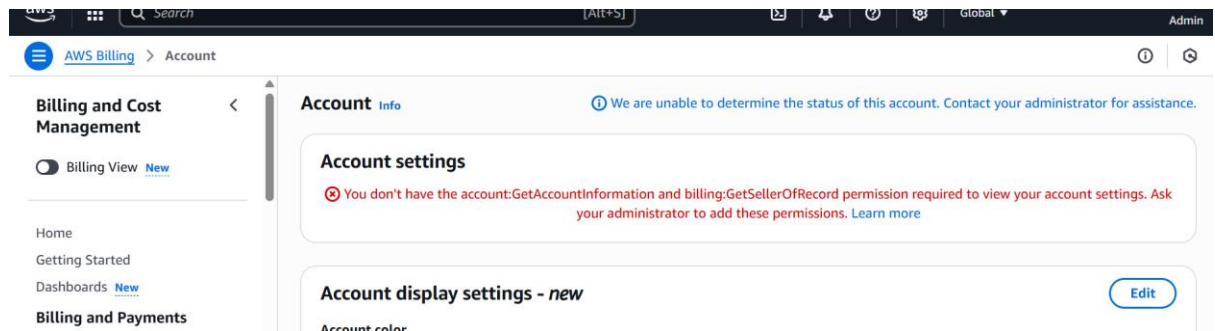
☐ Show Password

[Having trouble?](#)

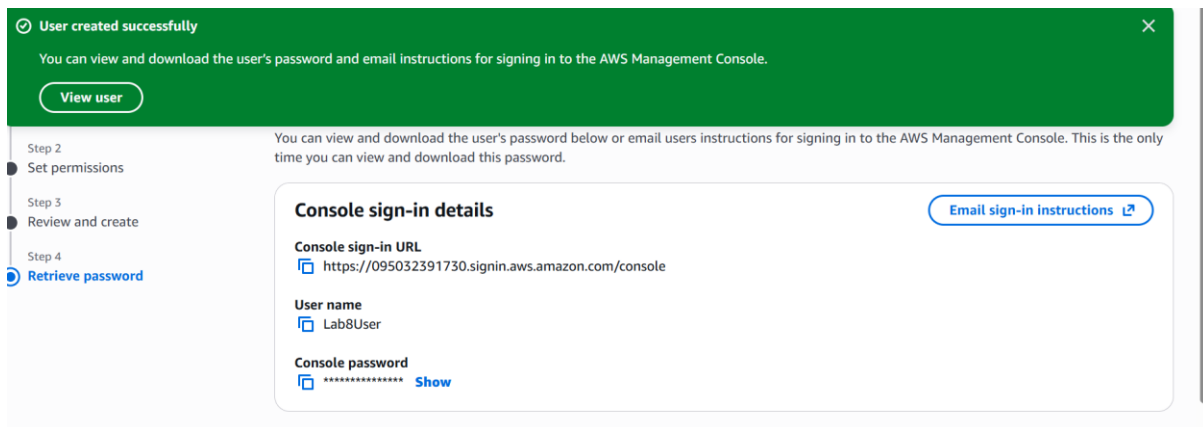
Sign in

Sign in using root user email

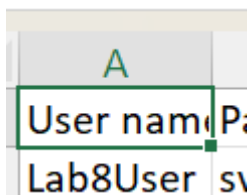
[Create a new AWS account](#)

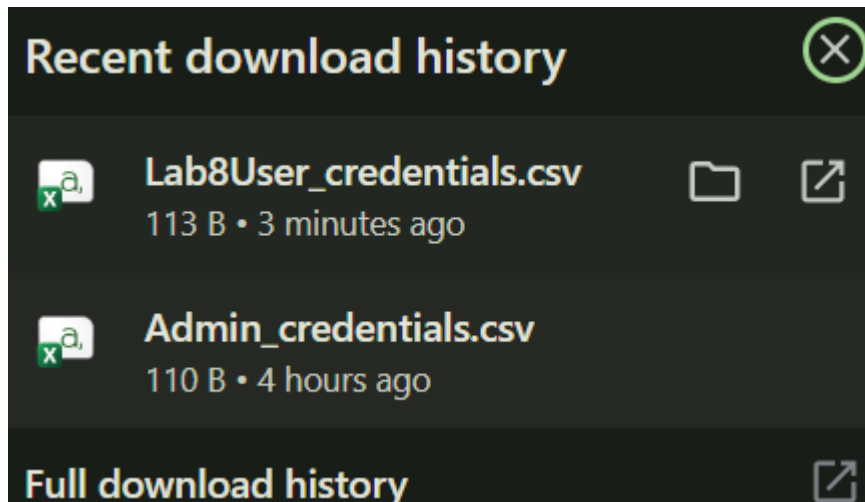


5. While logged in as Admin, create Lab8User:

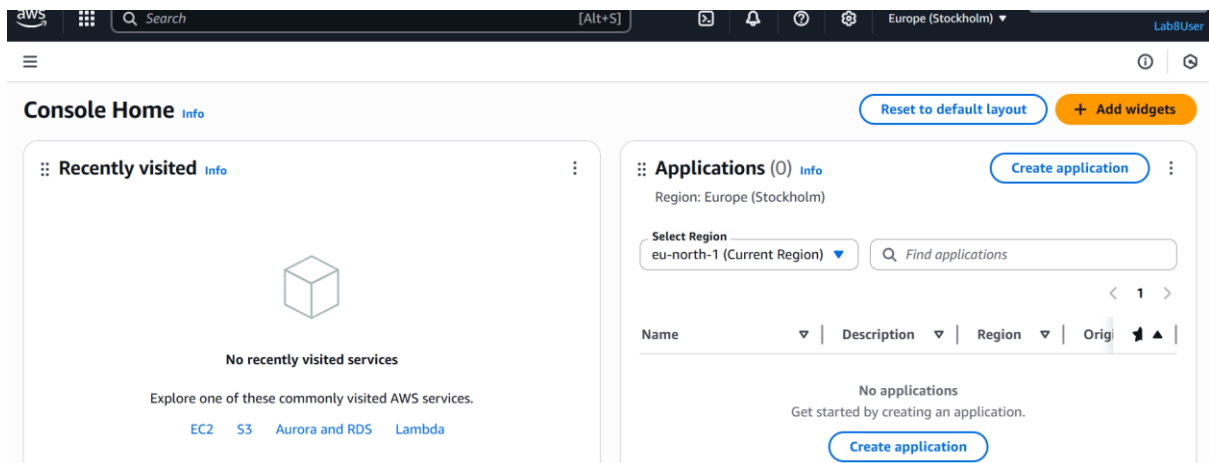


6. Download/save the Lab8User CSV on your Windows host (do not show password).

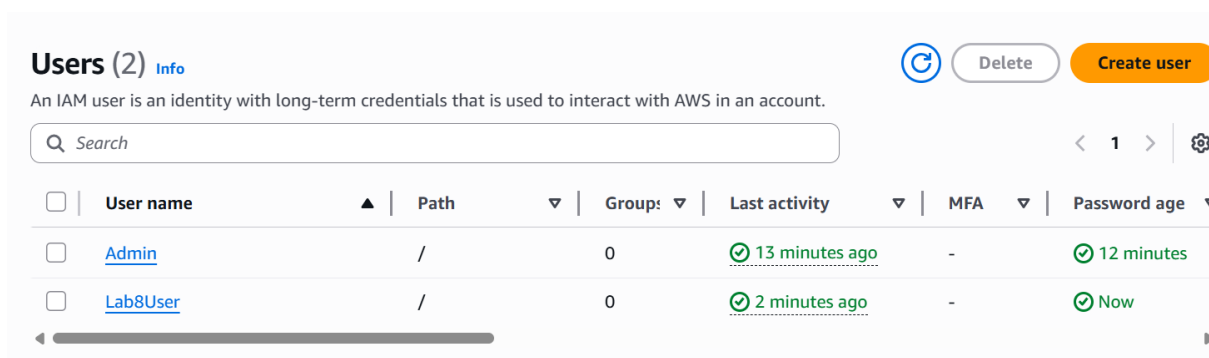




7. Logout Admin and login as Lab8User (use the Lab8User signin URL and credentials).
Capture after login:

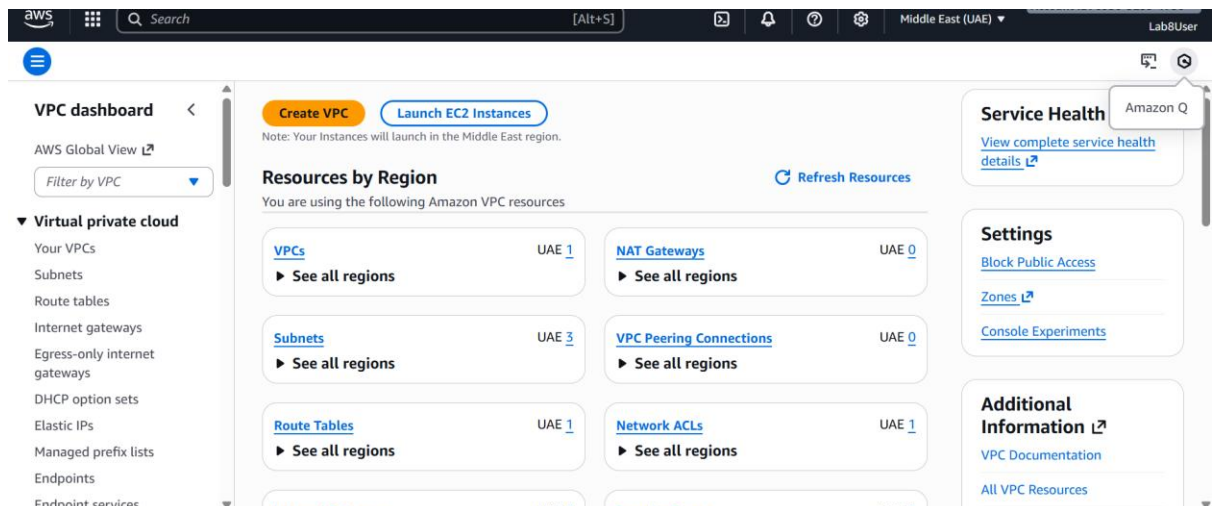


8. Task 2 summary (combine evidence):

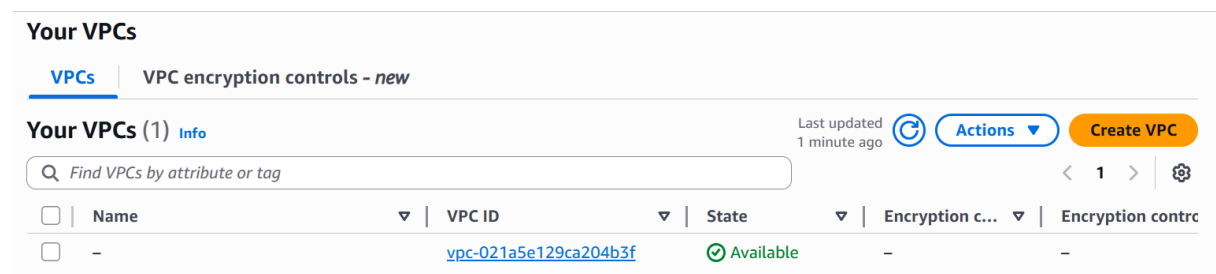


Task 3

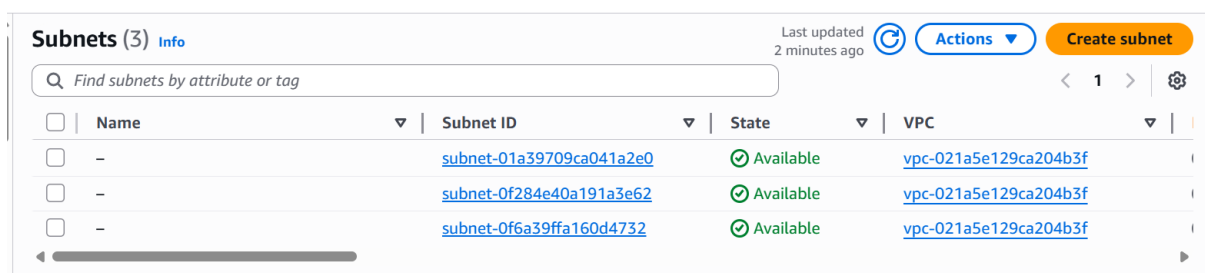
1. Open VPC console (Alt+S → "VPC") while region is me-central-1.



2. View VPCs list. Capture:



3. View Subnets list. Capture:



4. View Route Tables list. Capture:

Route tables (1) [Info](#)

Last updated 2 minutes ago

Actions

Create route table

Find route tables by attribute or tag

< 1 >

⚙️

☐	Name	Route table ID	Explicit subnet associ...	Edge associations	Main
☐	-	rtb-01d7dcd9b4e75165	-	-	Yes

Details [Info](#)

Route table ID

[rtb-01d7dcd9b4e75165](#)

Main

[Yes](#)

Explicit subnet associations

-

Edge associations

-

VPC

[vpc-021a5e129ca204b3f](#)

Owner ID

[095032391730](#)

Routes

Subnet associations

Edge associations

Route propagation

Tags

Routes (2)

Both

Edit routes

Filter routes

< 1 > ⚙️

Destination	Target	Status	Propagated	Route Origin
0.0.0.0/0	igw-0afceba38fb00c557	Active	No	Create Route
172.31.0.0/16	local	Active	No	Create Route Table

5. View Network ACLs list. Capture:

Network ACLs (1) [Info](#)

Actions

Create network ACL

Find Network ACLs by attribute or tag

< 1 >

⚙️

☐	Name	Network ACL ID	Associated with	Default	VPC ID
☐	-	acl-0ff9576f0aecd3ffc	3 Subnets	Yes	vpc-021a5e129

6. Task 3 summary (combine evidence):

aws

Search

[Alt+S]

Middle East (UAE)

Lab8User

VPC > Network ACLs

Egress-only internet gateways

DHCP option sets

Elastic IPs

Managed prefix lists

Endpoints

Endpoint services

NAT gateways

Peering connections

Route servers

Security

Network ACLs

Security groups

DNS firewall

Rule groups

Network ACLs (1) [Info](#)

Actions

Create network ACL

Find Network ACLs by attribute or tag

< 1 >

⚙️

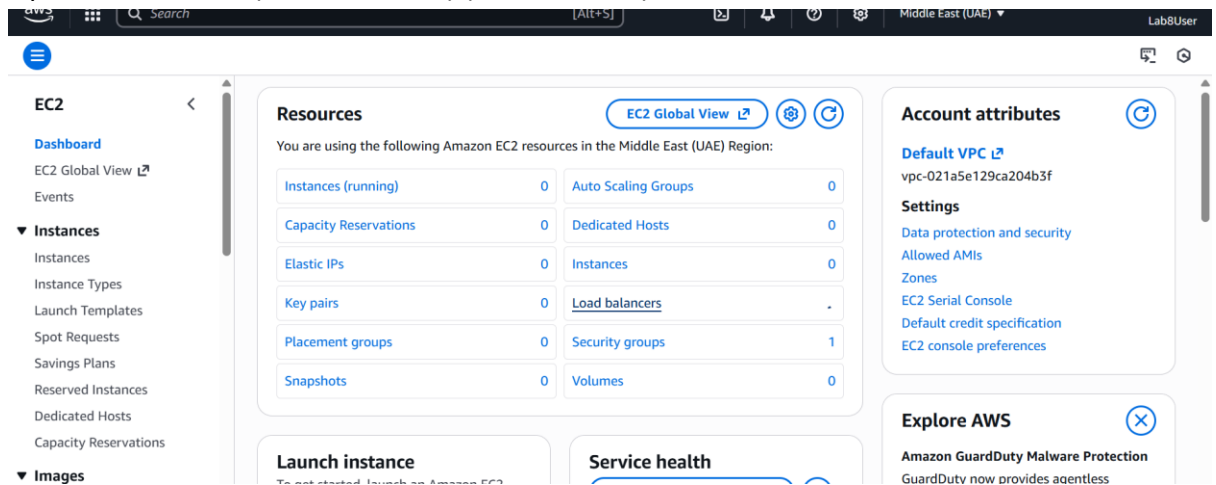
☐	Name	Network ACL ID	Associated with	Default	VPC ID
☐	-	acl-0ff9576f0aecd3ffc	3 Subnets	Yes	vpc-021a5e129

Select a network ACL

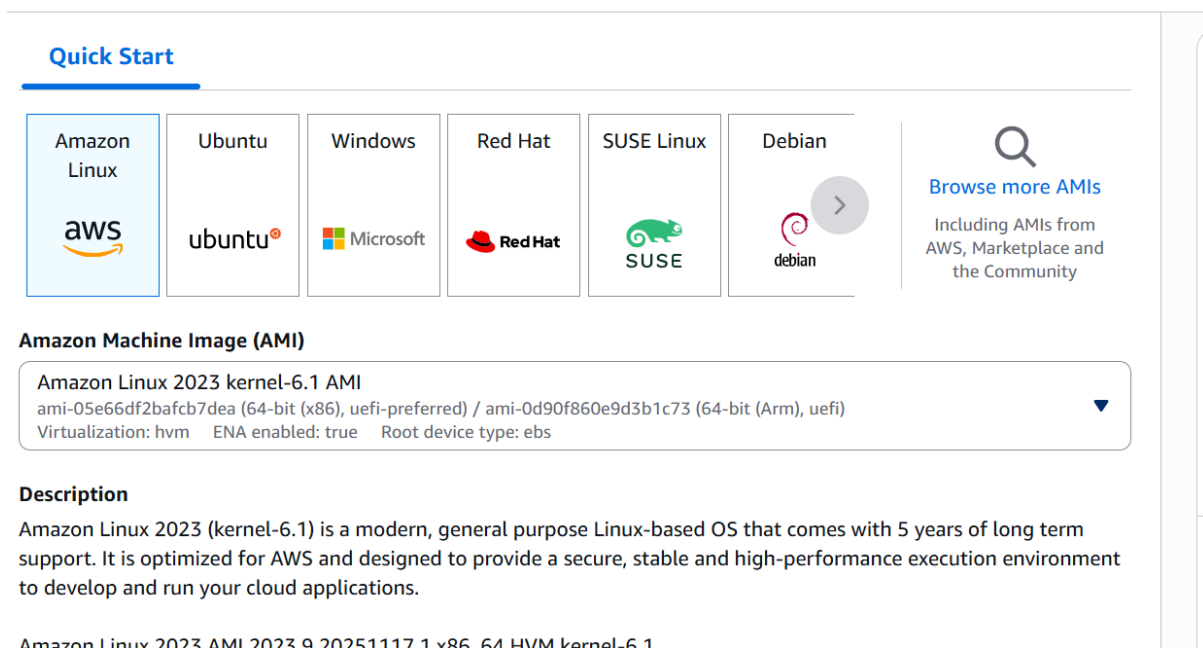
⚙️

Task 4

1. Open EC2 Console (Alt+S → "EC2") (me-central-1).



2. Instance Launch configuration (during review before launching). Configure:



▼ Instance type [Info](#) | [Get advice](#)

Instance type

t3.micro

Family: t3 2 vCPU 1 GiB Memory Current generation: true

On-Demand Ubuntu Pro base pricing: 0.016 USD per Hour

On-Demand Linux base pricing: 0.0125 USD per Hour

On-Demand RHEL base pricing: 0.0413 USD per Hour

On-Demand SUSE base pricing: 0.0125 USD per Hour

On-Demand Windows base pricing: 0.0217 USD per Hour

☒ All generations

[Compare instance types](#)

Additional costs apply for AMIs with pre-installed software

▼ Key pair (login) [Info](#)

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - *required*

Lab8Key

 [Create new key pair](#)

VPC - *required* | [Info](#)

vpc-021a5e129ca204b3f
172.31.0.0/16

(default) ▼



Subnet | [Info](#)

No preference ▼



[Create new subnet](#)

Availability Zone | [Info](#)

No preference ▼



[Enable additional](#)

Auto-assign public IP | [Info](#)

Enable ▼

Firewall (security groups) | [Info](#)

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

☒ Create security group

☐ Select existing security group

Security group name - *required*

Lab8SecurityGroup

Inbound Security Group Rules

▼ Security group rule 1 (TCP, 22, 103.147.87.115/32)

Remove

Type [Info](#)

ssh

Protocol [Info](#)

TCP

Port range [Info](#)

22

Source type [Info](#)

My IP

Name [Info](#)

Q Add CIDR, prefix list or security

103.147.87.115/32 X

Description - optional [Info](#)

e.g. SSH for admin desktop

Add security group rule

▼ Configure storage [Info](#)

Advanced

1x

8

GiB

gp3

Root volume, 3000 IOPS, Not encrypted

Add new volume

🕒 Click refresh to view backup information



The tags that you assign determine whether the instance will be backed up by any Data Lifecycle Manager policies.

0 x File systems

Edit

► Advanced details [Info](#)

📄 Lab8Key.pem

12/7/2025 5:17 PM

PEM File

🟢 Success

Successfully initiated launch of instance (i-0f4a9ce71ad539bc3)

► Launch log

Next Steps

Q What would you like to do next with this instance, for example "create alarm" or "create backup"

< 1 2 3 4 5 6 >

3. After launch, EC2 Instances list showing Lab8Machine in "running" state and public IPv4 visible.

EC2 > Instances > i-0f4a9ce71ad539bc3

EC2

- Dashboard
- EC2 Global View
- Events
- Instances**
 - Instances
 - Instance Types
 - Launch Templates
 - Spot Requests
 - Savings Plans
 - Reserved Instances
 - Dedicated Hosts
 - Capacity Reservations
- Images

Instance summary for i-0f4a9ce71ad539bc3 (Lab8Machine)

Updated less than a minute ago

[Connect](#) [Instance state](#) [Actions](#)

Instance ID i-0f4a9ce71ad539bc3	Public IPv4 address 51.112.244.212 open address	Private IPv4 addresses 172.31.5.242
IPv6 address -	Instance state Running	Public DNS ec2-51-112-244-212.me-central-1.compute.amazonaws.com open address
Hostname type IP name: ip-172-31-5-242.me-central-1.compute.internal	Private IP DNS name (IPv4 only) ip-172-31-5-242.me-central-1.compute.internal	Elastic IP addresses -
Answer private resource DNS name IPv4 (A)	Instance type t3.micro	AWS Compute Optimizer finding -
Auto-assigned IP address	VPC ID	

- On Windows host, run SSH using the downloaded .pem (PowerShell/Git Bash/Windows Terminal):

```
ssh -i <path>/Lab8Key.pem ec2-user@<public-IP>
```

Capture the SSH command and successful shell prompt on the EC2 instance:

```
PS C:\Users\HABIB TARIQ> ssh -i "C:\Users\HABIB TARIQ\Downloads\Lab8Key.pem" ec2-user@3.29.127.228
The authenticity of host '3.29.127.228 (3.29.127.228)' can't be established.
ED25519 key fingerprint is SHA256:ApHPyVf7vb7eDmpXWPvzrrgTtp/e40cgAbvLx4X7QXHk.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '3.29.127.228' (ED25519) to the list of known hosts.

#_
#####
~\#####
~~\#####\
~~\###|
~~\#/---
~~V~'t-->
    https://aws.amazon.com/linux/amazon-linux-2023

[ec2-user@ip-10-0-23-103 ~]$
```

- Run the install commands on the EC2 shell:

10. Save and verify file exists:

```
[ec2-user@ip-10-0-23-103 ~]$ sudo vim compose.yaml
```

11. Save screenshot as: task4_compose_yaml_saved_ls.png — ls -l showing compose.yaml present.

```
[ec2-user@ip-10-0-23-103 ~]$ ls -l
total 4
-rw-r--r--. 1 root root 1129 Dec 30 08:36 compose.yaml
[ec2-user@ip-10-0-23-103 ~]$
```

12. Add ec2-user to docker group, show groups before re-login, exit and reconnect, show groups after reconnect:

```
groups # user does not docker permission
```

```
sudo usermod -aG docker $USER
```

```
groups    # before re-login
```

exit

```
[ec2-user@ip-10-0-23-103 ~]$ groups
ec2-user adm wheel systemd-journal
[ec2-user@ip-10-0-23-103 ~]$ sudo usermod -aG docker $USER
[ec2-user@ip-10-0-23-103 ~]$ groups
ec2-user adm wheel systemd-journal
[ec2-user@ip-10-0-23-103 ~]$ exit
logout
Connection to 3.29.127.228 closed.
PS C:\Users\HARIB TAP\...>
```

13. # Reconnect

```
ssh -i <path>/Lab8Key.pem ec2-user@<public-IP>
```

```
Connection to 3.29.127.228 closed.  
PS C:\Users\HABIB TARIQ> ssh -i "C:\Users\HABIB TARIQ\Downloads\Lab8Key.pem" ec2-user@3.29.127.228  
  
      #_  
    ~ \_ #####  
  NN   \_#####\  
  NN     \####|  
  NN       \|#/_____  
  NN        V~!  !->  
  NNNN  
  NN  _-_-_  
    _/_/_/_/  
   /m/'
```

Amazon Linux 2023

<https://aws.amazon.com/linux/amazon-linux-2023>

Last login: Tue Dec 30 08:45:06 2025 from 103.147.87.212
[ec2-user@ip-10-0-23-103 ~]\$ |

14. groups # after re-login (should include docker)

```

Last login: Tue Dec 30 08:43:00 2020 from 10
[ec2-user@ip-10-0-23-103 ~]$ groups
ec2-user adm wheel systemd-journal docker
[ec2-user@ip-10-0-23-103 ~]$ |

```

15. Run docker compose up -d from the directory with compose.yaml:

docker compose up -d

```

[+] up 16/18tgres:alpine [#####] 108.8MB / 112MB Pulling
[+] up 16/18tgres:alpine [#####] 108.8MB / 112MB Pulling
[+] up 16/18tgres:alpine [#####] 108.8MB / 112MB Pulling
[+] up 16/18tgres:alpine [#####] 108.8MB / 112MB Pulling
[+] up 16/18tgres:alpine [#####] 111.9MB / 112MB Pulling
[+] up 16/18tgres:alpine [#####] 111.9MB / 112MB Pulling
* Image postgres:alpine [#####] 112MB / 112MB Pulling
[+] up 23/23eec0d Pull complete 4.3s
✓ Image postgres:alpine Pulled 16.8s
  ✓ 1074353eec0d Pull complete 4.3s
  ✓ 51a9324a2bdc Pull complete 5.0s
  ✓ 1e827ac0fa1e Pull complete 5.7s
  ✓ 0629ac8b8cc5 Pull complete 6.1s
  ✓ 2e50a444bddf Pull complete 13.5s
  ✓ 4465107e1675 Pull complete 13.5s
  ✓ cc2ee20b6816 Pull complete 13.5s
  ✓ 0dfd86289dd1 Pull complete 13.6s
  ✓ 98eccc0da868 Pull complete 13.6s
  ✓ b747867e61cb Pull complete 13.6s
✓ Image gitea/gitea:latest Pulled 9.5s
  ✓ 2d35ebdb57d9 Pull complete 2.1s
  ✓ 9f4e672c1f34 Pull complete 3.2s
  ✓ 4da94b3cc809 Pull complete 3.2s
  ✓ 8e016ece0bd3 Pull complete 3.2s
  ✓ 346e493cb6cb Pull complete 6.2s
  ✓ 95215379f1d1 Pull complete 6.4s
✓ Network webnet Created 0.1s
✓ Volume gitea Created 0.0s
✓ Volume gitea_postgres Created 0.0s
✓ Container gitea_db Created 0.1s
✓ Container gitea Created 0.1s
[ec2-user@ip-10-0-23-103 ~]$ |

```

16. Edit the security group Lab8SecurityGroup inbound rules in the EC2 console: add Custom TCP rule port 3000 source 0.0.0.0/0 and save. Capture the inbound rules after saving:

Inbound rules [Info](#)

Inbound rule 1

Delete

Security group rule ID
sgr-0d66db7078abd5e6b

Type [Info](#)

Custom TCP

Protocol [Info](#)

TCP

Port range [Info](#)

3000

Source type [Info](#)

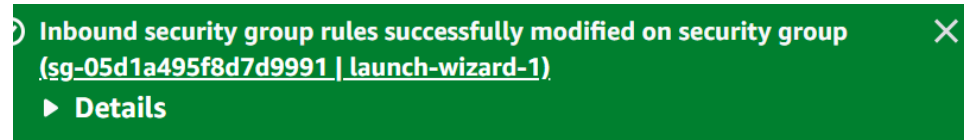
Custom

Source [Info](#)

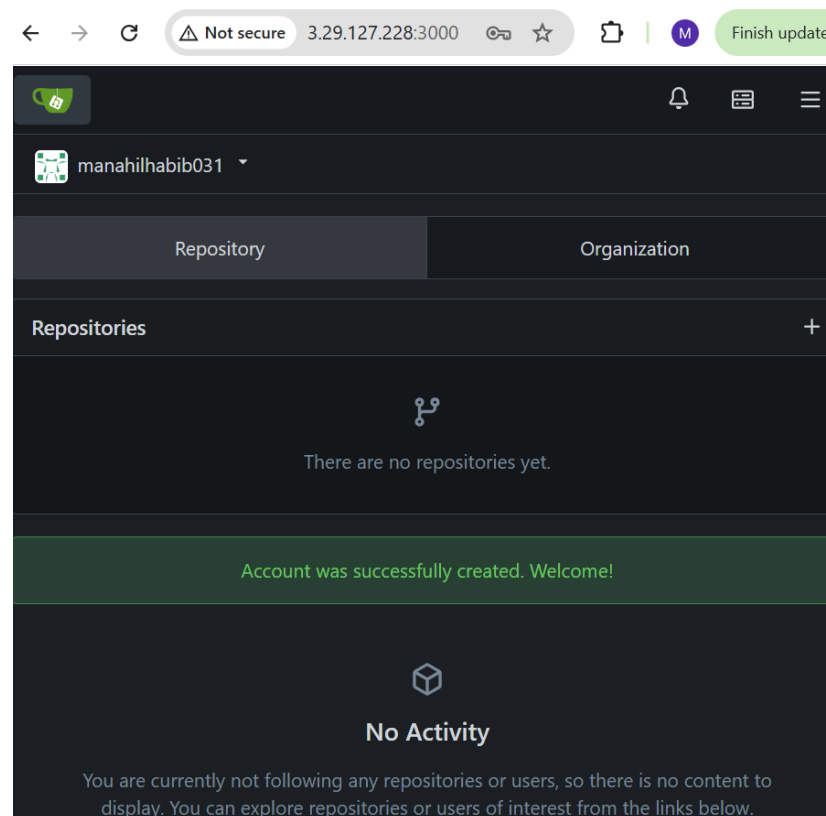
Q

0.0.0.0/0

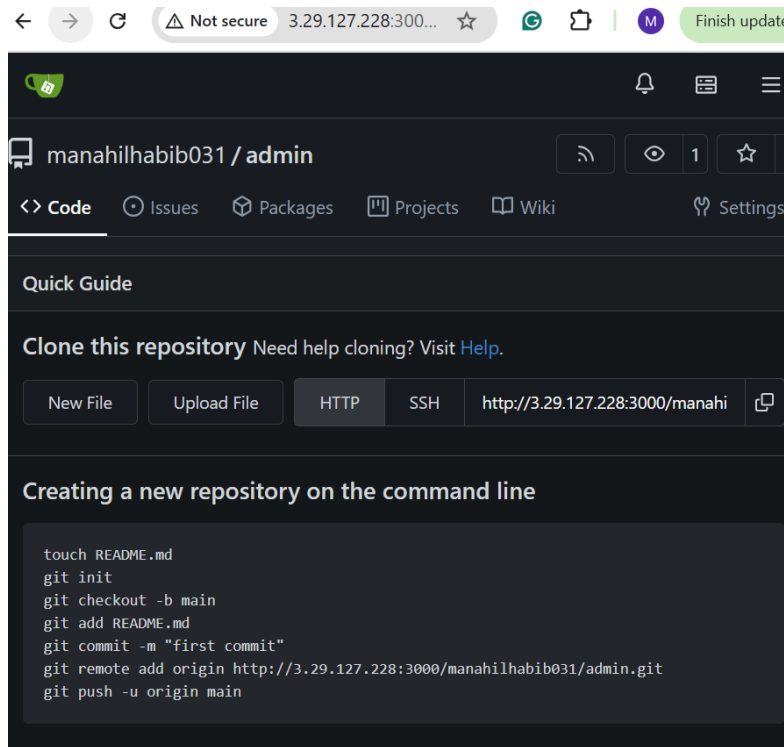
Description - optional [Info](#)



17. From your Windows browser navigate to: <http://Public-IP:3000> — capture the Gitea setup/install page:



18. Complete initial Gitea setup (create admin user, create a repo) and capture Gitea showing the created repository:



Cleanup — Remove resources to avoid charges

1. Terminate the EC2 instance Lab8Machine.

Terminate (delete) instance

 On an EBS-backed instance, the default action is for the root EBS volume to be deleted when the instance is terminated. Storage on any local drives will be lost.

Are you sure you want to terminate these instances?

Instance ID	Termination protection
 i-05c7874fbcd6a7388 (Lab8Machine)	 Disabled

To confirm that you want to delete the instances, choose the terminate button below. Instances with termination protection enabled will not be terminated. Terminating the instance cannot be undone.

Skip OS shutdown

This option skips the graceful OS shutdown process. Use only when your instance must be stopped immediately, such as during an emergency or failover.

☐ Skip OS shutdown

Cancel

Terminate (delete)

☐	Lab8Machine	i-0f4a9ce71ad539bc3	 Terminated  	t3.micro	
--------------------------	-------------	---------------------	--	----------	---

2. Delete security group Lab8SecurityGroup and key pair Lab8Key from the EC2 console (after instances terminated).

☐	Name	Volume ID	Type
You currently have no volumes in this region			
1			

Key pairs [Info](#)



Actions

Create key pair

Find Key Pair by attribute or tag

< 1 >

☐	Name	Type	Created	Fingerprint
--------------------------	------	------	---------	-------------

No key pairs to display

Delete security groups



Are you sure that you want to delete this security group?

- sg-06a922f201453a797 - Lab8SecurityGroup

Cancel

Delete

- Delete IAM users Lab8User and any access keys.

User "Lab8User" deleted.



- Final cleanup summary (show billing or resource groups with no active resources if possible).

Users (0) [Info](#)



Delete

Create user

An IAM user is an identity with long-term credentials that is used to interact with AWS in an account.

Search

< 1 >

☐	User name	Path
--------------------------	-----------	------
